

Financial Time-Series Forecasting & Risk Analysis

Indian Stocks and NIFTY 50 | Forecasting, Portfolio Analytics and Risk Modelling

Select Asset

HDFC Bank

735.00

Latest Close

23.39%

Total Return

3.24%

CAGR

25.66%

Annual Volatility

-0.11

Sharpe Ratio

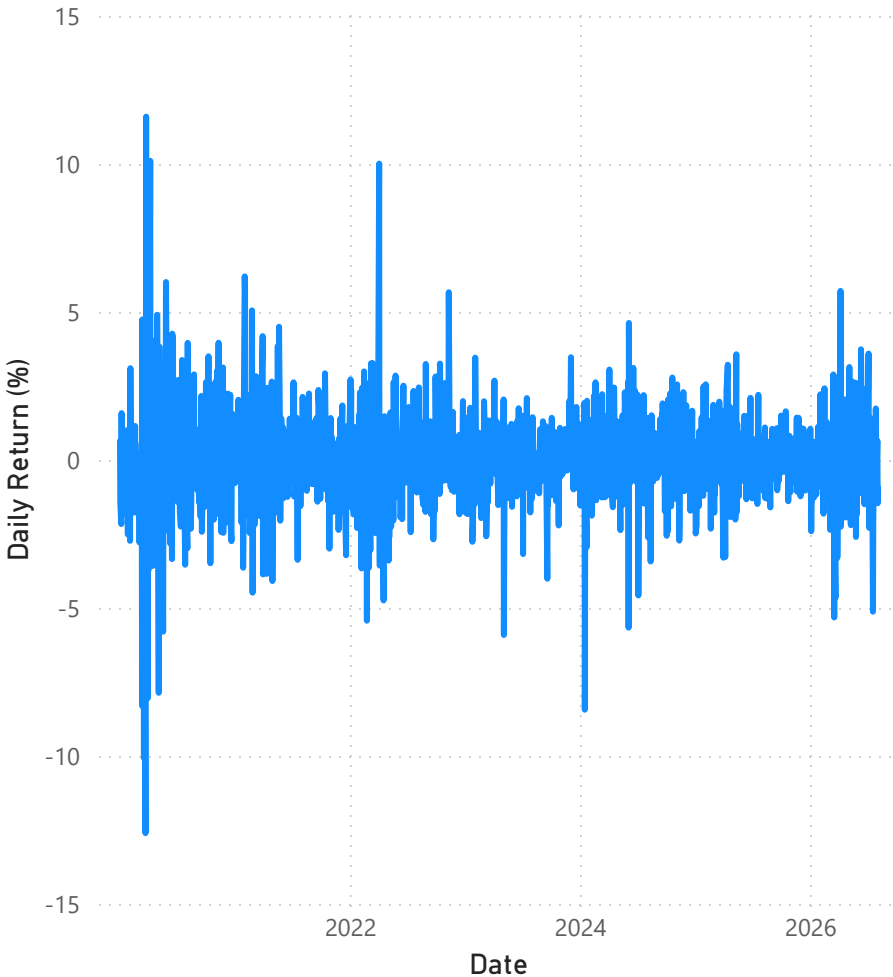
-40.47%

Maximum Drawdown

Historical Adjusted Close Trend



Daily Return Fluctuations (%)



Portfolio & Risk Analysis

Equal-Weight Portfolio | NIFTY 50 Benchmark, VaR and Monte Carlo Simulation

Market Overview

Portfolio & Risk

Forecasting

Regression & Correlation

10,00,000.00

Portfolio Value (INR)

17,075.35

Historical VaR (95%)

28,568.86

Historical CVaR (95%)

19,975.45

Parametric VaR (95%)

92,649.97

Monte Carlo VaR (30-Day)

42.34%

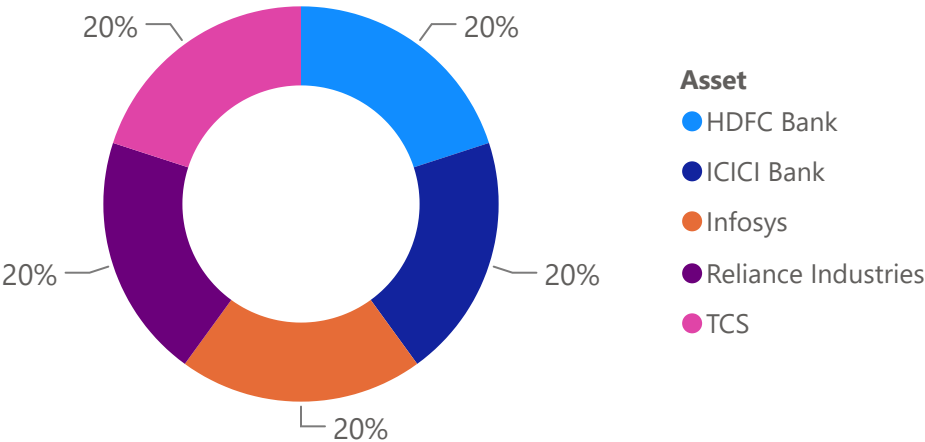
Probability of Loss

Note: VaR and CVaR values are shown in INR.

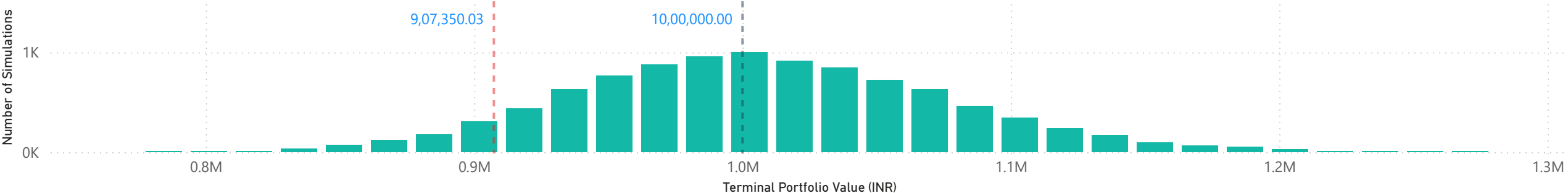
Equal-Weight Portfolio vs NIFTY 50



Equal-Weight Portfolio Allocation



Distribution of 30-Day Simulated Portfolio Values



NIFTY 50 Forecasting & Model Comparison

ARIMA vs Prophet | 60-Day Holdout Evaluation and 30-Business-Day Forecast

ARIMA(0, 1, 0)

Best Model

2.26%

ARIMA MAPE

7.14%

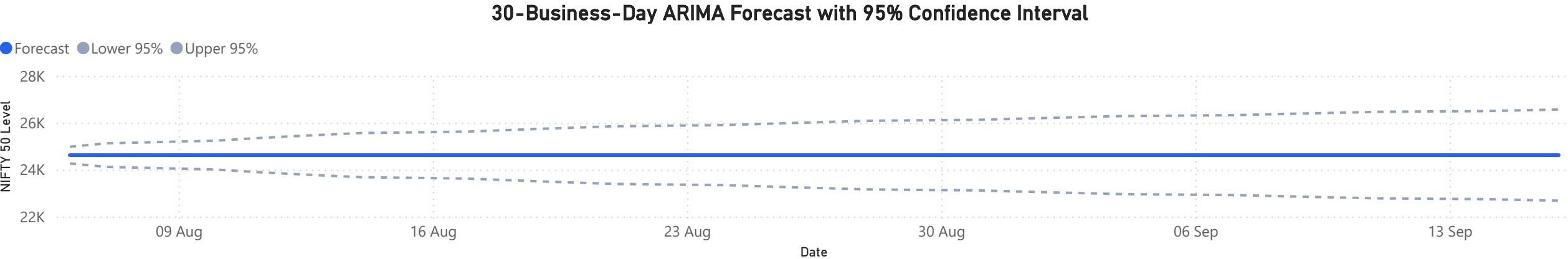
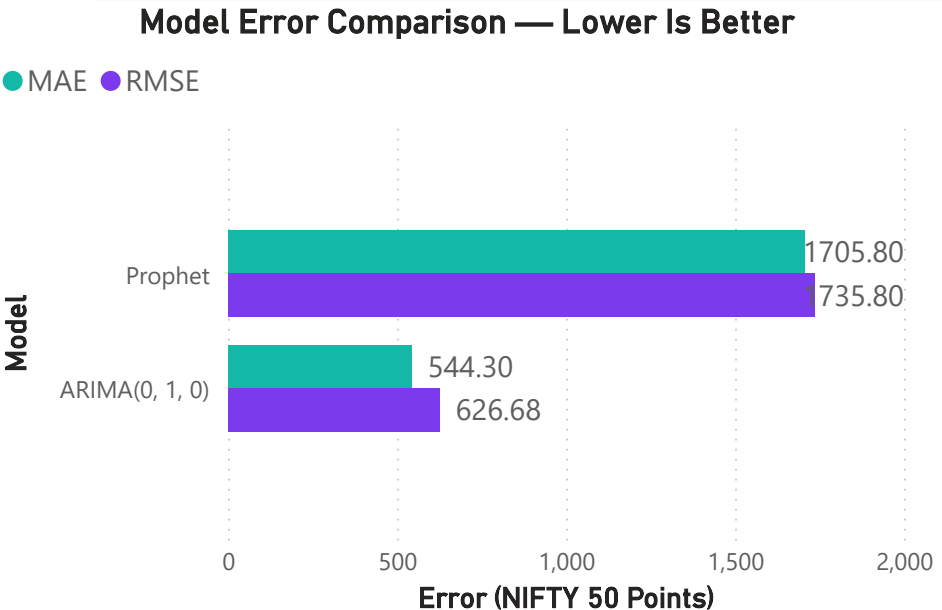
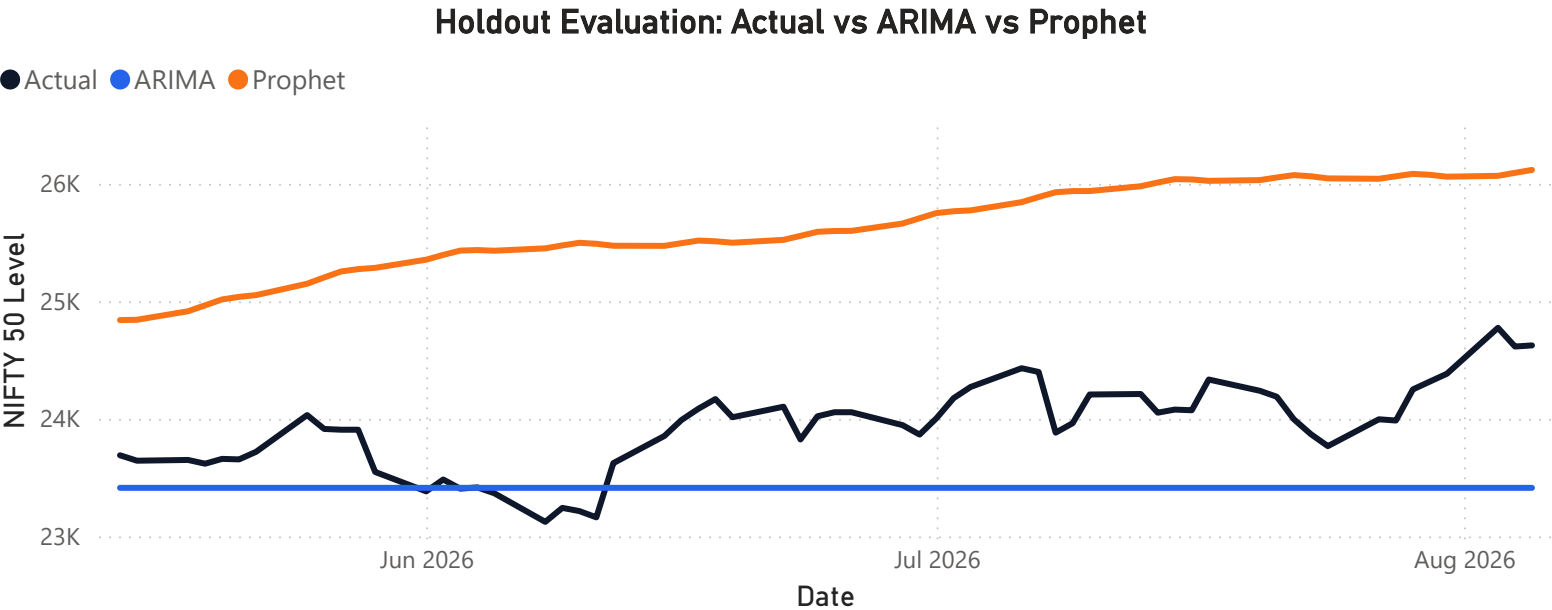
Prophet MAPE

626.68

ARIMA RMSE

1,735.80

Prophet RMSE



Regression & Correlation Analysis

Market Sensitivity, Alpha, R-Squared and Daily Return Relationships

Market Overview

Portfolio & Risk

Forecasting

Regression & Correlation

1.09

Beta

-7.03%

Annualised Alpha

57.58%

R-Squared

16.71%

Residual Volatility

-0.0279%

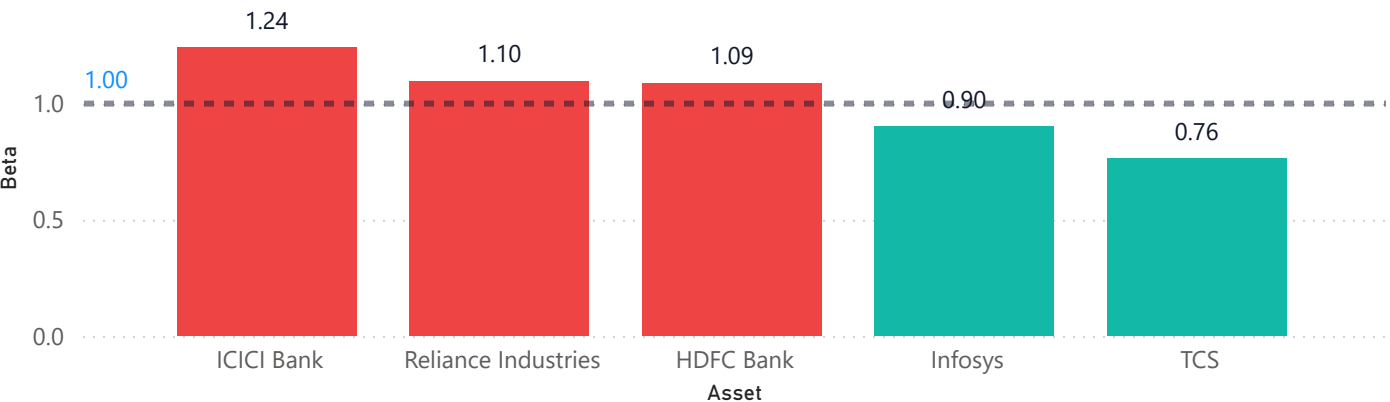
Daily Alpha

Select Asset

HDFC Bank



Stock Beta Relative to NIFTY 50



Correlation Matrix of Daily Returns

Asset 1	HDFC Bank	ICICI Bank	Infosys	NIFTY 50	Reliance Industries	TCS
HDFC Bank	1.00	0.63	0.33	0.76	0.44	0.29
ICICI Bank	0.63	1.00	0.34	0.76	0.42	0.30
Infosys	0.33	0.34	1.00	0.57	0.33	0.73
NIFTY 50	0.76	0.76	0.57	1.00	0.71	0.55
Reliance Industries	0.44	0.42	0.33	0.71	1.00	0.35
TCS	0.29	0.30	0.73	0.55	0.35	1.00

HDFC Bank Daily Return vs NIFTY 50 Return

